FDS EXPERIMENT-14

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Experiment to understand feature scaling.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.preprocessing import MinMaxScaler
from sklearn.preprocessing import StandardScaler
```

```
data = {
    "Age": [18, 20, 22, 24, 25],
    "Study_Hours": [2, 4, 6, 8, 10],
    "Marks": [40, 50, 65, 80, 95]
}

df = pd.DataFrame(data)

print(df)
```

```
   Age  Study_Hours  Marks
0   18            2     40
1   20            4     50
2   22            6     65
3   24            8     80
4   25           10     95
```

```python
#Min - Max Normalization
scaler = MinMaxScaler()

normalized_data = scaler.fit_transform(df)

normalized_df = pd.DataFrame(
    normalized_data,
    columns=df.columns
)

print(normalized_df.round(2))
```

```
    Age  Study_Hours  Marks
0  0.00         0.00   0.00
1  0.29         0.25   0.18
2  0.57         0.50   0.45
3  0.86         0.75   0.73
4  1.00         1.00   1.00
```

```python
#Z - score normalization
scaler = StandardScaler()

standardized_data = scaler.fit_transform(df)

standardized_df = pd.DataFrame(
    standardized_data,
    columns=df.columns
)

print(standardized_df.round(2))
```

```
    Age  Study_Hours  Marks
0 -1.48        -1.41  -1.31
1 -0.70        -0.71  -0.81
2  0.08         0.00  -0.05
3  0.86         0.71   0.71
4  1.25         1.41   1.46
```